ML TRAINING - WEEK 2

CALIFORNIA HOUSING PRICE PREDICTOR

Deep learning model tracked with MLflow, served with FastAPI + Docker, and used through a Streamlit app.

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Data Science | October 2026



PREDICT | PACKAGE | DEPLOY

THE DATA

Dataset & Preparation

20,640
districts in the California housing data

13
model features (8 numeric + 5 location)

207
missing values in total_bedrooms

$
target: median_house_value

DATA PIPELINE

1 Clean
Filled 207 missing total_bedrooms with the median

2 Encode
One-hot encoded ocean_proximity into 5 columns

3 Split
80/20 train/test split
16,512 train | 4,128 test

4 Scale
StandardScaler fitted on the training data, saved as scaler.pkl

THE MODELS

Three Neural Networks, Six Experiments

Baseline MLP

Simple starting point

- Dense 64 (ReLU)
- Dense 32 (ReLU)
- Output 1

Deeper DNN

More layers, more capacity

- Dense 128 (ReLU)
- Dense 64 (ReLU)
- Dense 32 (ReLU)
- Dense 16 (ReLU)
- Output 1

Regularised DNN

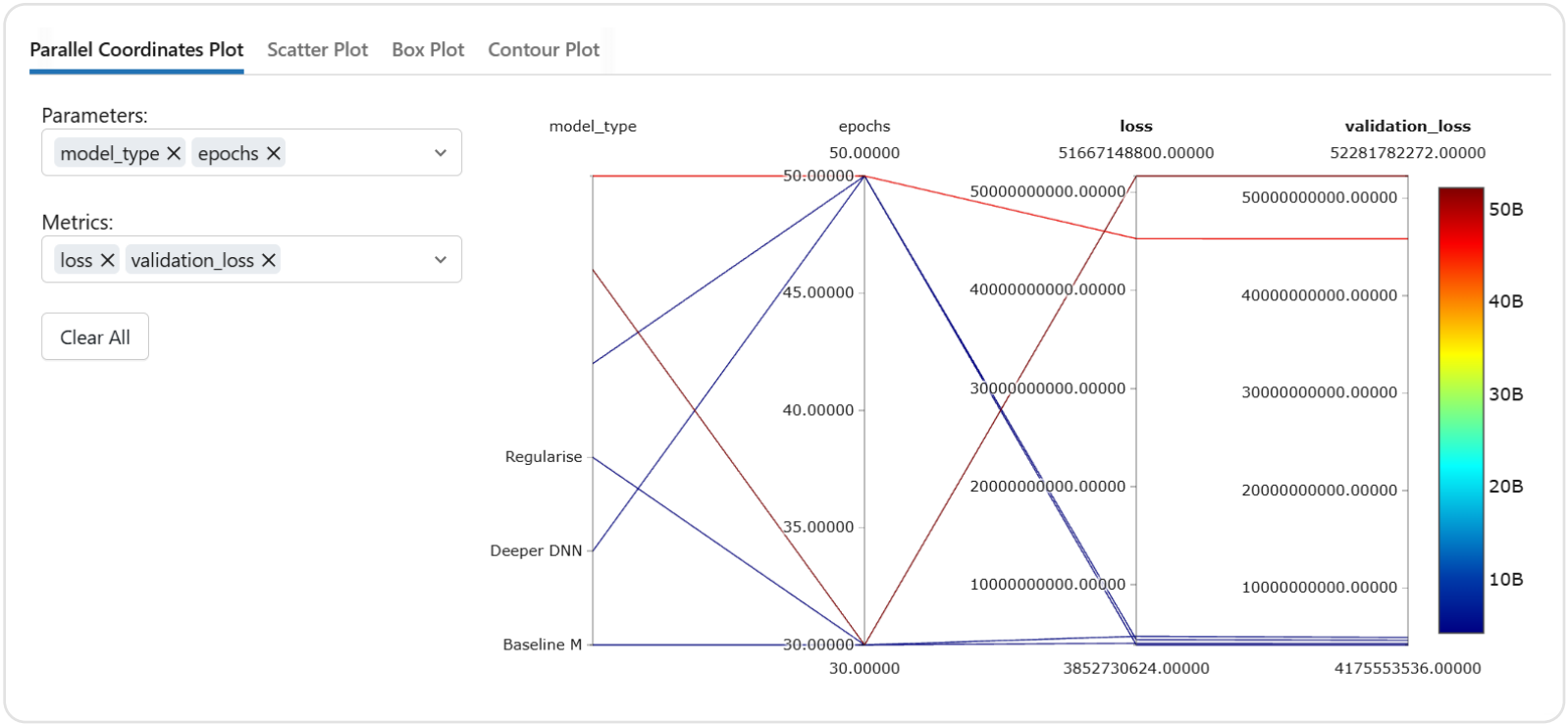
BatchNorm + Dropout

- Dense 64 + BatchNorm
- Dropout 0.1
- Dense 32 + BatchNorm
- Dropout 0.2
- Output 1

OPTIMIZER	LOSS	METRIC	EPOCHS	BATCH SIZE	VALIDATION
Adam	Mean squared error	MAE	30 and 50	30	20% of train

EXPERIMENT TRACKING

Every Run Tracked in MLflow



Parallel coordinates view: model type and epochs against loss and validation loss.

WHAT WE TRACKED

6 runs in one experiment (w2-syed-taha)

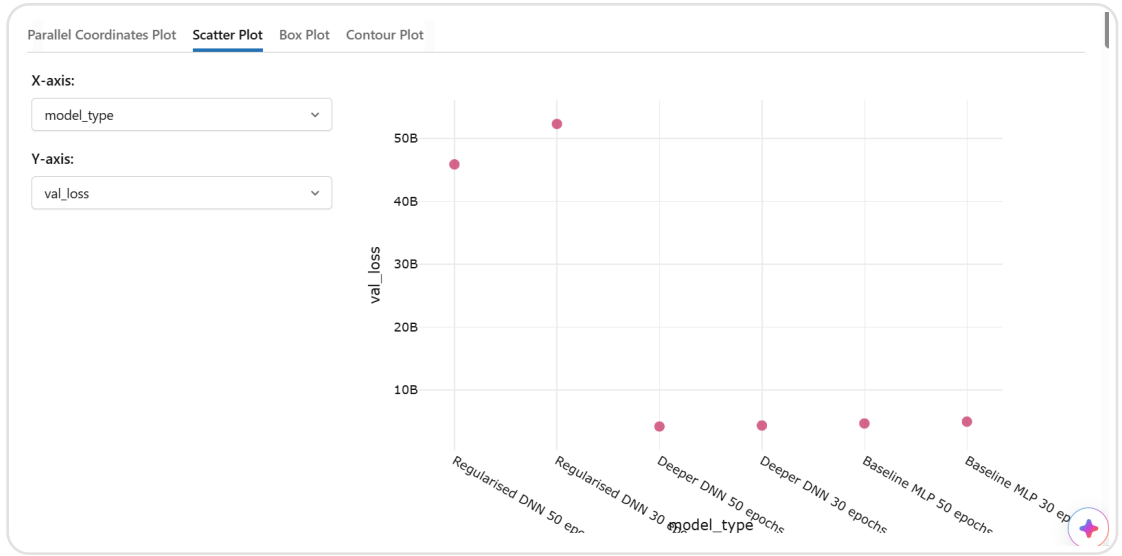
2 parameters: model_type, epochs

3 metrics: loss, val_loss, val_mae

1 best model saved automatically

RESULTS

Deeper DNN Wins



Validation loss by run (lower is better).

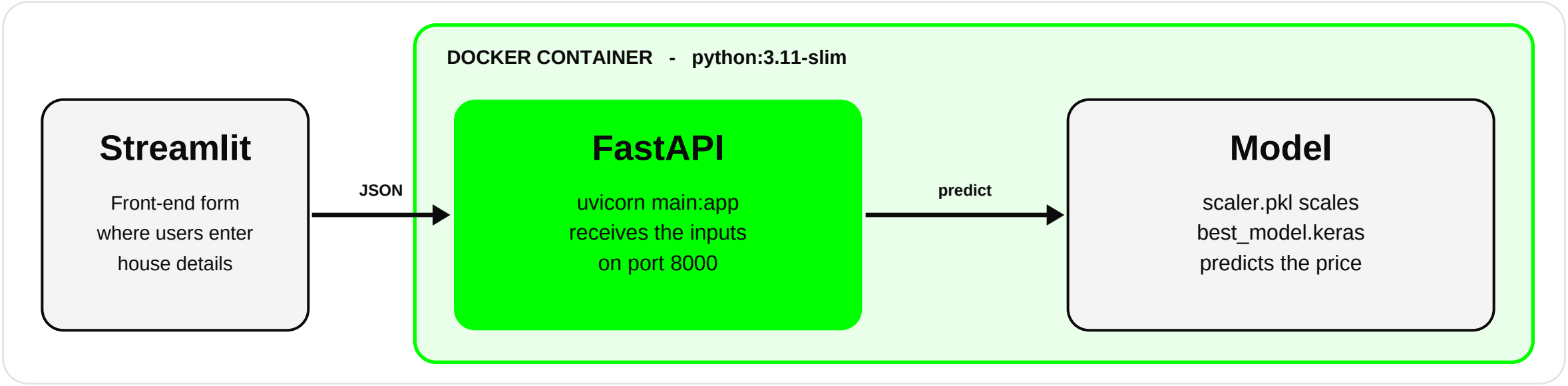
VALIDATION LOSS (MSE, BILLIONS) | VAL MAE ($)

Deeper DNN, 50 epochs	4.18B	**$44.9K**
Deeper DNN, 30 epochs	4.34B	**$46.1K**
Baseline MLP, 50 epochs	4.66B	**$48.3K**
Baseline MLP, 30 epochs	4.95B	**$49.8K**
Regularised, 50 epochs	45.85B	**$188.4K**
Regularised, 30 epochs	52.28B	**$200.3K**

BEST MODEL	VAL LOSS	VAL MAE	VAL RMSE
Deeper DNN, 50 epochs	4.18 billion	~$44.9K	~$64.6K

DEPLOYMENT

From Model to a Running API



RUN IT

```
$ docker build -t taha_image .
$ docker run -d -p 8000:8000 --name taha_container taha_image
```

Same image runs the same way on any machine. Interactive API docs are available at /docs.

THE FRONT-END

Streamlit App



1 **Enter details**

Longitude, latitude, housing age, rooms, bedrooms, population and more.

2 **One click**

The app sends the values to the FastAPI service running in Docker.

3 **Instant price**

The model returns the predicted median house value.

WRAP UP

Key Takeaways & Next Steps

WHAT WE BUILT

1. Cleaned and scaled 20,640 housing records
2. Compared 3 architectures x 2 epoch settings
3. Tracked all 6 runs with MLflow
4. Picked the best model: Deeper DNN, 50 epochs
5. Served it with FastAPI inside Docker
6. Built a Streamlit app for end users

NEXT STEPS

Train longer

Regularised models were still improving at 50 epochs.

Tune it

Try learning rates, dropout and layer sizes.

Test set

Report final results on the held-out test data.

Go live

Deploy the Docker image to the cloud.